COMPUTER ENGINEERING

QUEZON CITY

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SEATWORK NO. 7.1 Using Sorting Algorithms

1. PROCEDURE OUTPUT

What procedures did you perform?

1. What is a sorting algorithm?

A sorting algorithm is a fundamental concept of arranging a given list into a specific order. It is done through different sorting process; with each having their own advantages in a given set of data. A sorting algorithm is commonly used (but not limited to) to arrange a list of data in a numerical or alphabetical manner depending on the conditions. A sorting algorithm uses comparisons and iterations like in our previous topic in order to rearrange a certain data. A sorting algorithm can be evaluated via time-complexity which is our previous topic, and space-complexity.

2.Where can sorting algorithms be used?

Sorting algorithms can be used whenever a collection of data needs to be organized into a specific order. They are commonly applied to lists of numbers, words, names, cities, records, and other types of information. For example, data can be arranged alphabetically, by size, by value, by population, or by another selected criterion. Sorting is especially useful when working with large amounts of data, because an organized list can make other operations, such as searching and comparing information, more efficient. The choice of sorting method may also depend on the number of items being processed, since some methods are more suitable for small collections while others are designed to handle larger datasets efficiently.

3. Explain the different types of sorting algorithms.

Bubble Sort
- Makes repeated passes through the list, comparing adjacent items, and if they are not in order, it would swap them. In each pass, the largest value would "bubble" up to its correct position.

Selection Sort
- A better version of bubble sort, instead of making many exchanges it would only do one per pass. It scans the unsorted portion, to find either the largest or smallest value, swaps it into its correct position in one exchange, then scans the unsorted portion again.

Insertion Sort
- Compared to the first two sorting algorithms, each new item is picked up and "inserted" into the correct position within the already-sorted list, shifting larger items one slot to the right, rather than using a temporary variable

swap to make room for the item that would be inserted.

4. Give sample programs in C++ that use sorting algorithms, specifically selection sort, insertion sort, and bubble sort. Explain how the programs work.

Selection Sort
You look through your entire unsorted list, find the absolute smallest item, and move it to the front. Then, you search the rest of the list for the next smallest item and place it second. You repeat this selection process until everything is lined up. It's straightforward, but it takes the same amount of effort even if the list is already mostly sorted. For example, in a list [5, 2, 8,1], it scans all four numbers, finds 1 is the smallest and swaps it to the front, and does the same for the other numbers, getting [1, 2, 5, 8].

Insertion Sort
It is a sorting algorithm that inserts each element of an unsorted list into its proper position in a sorted list. It takes the second index element of the list and uses it as the first element to be sorted. Consider the array: [4, 2, 16, 8]. Then, it compares this element with the first index element if its value is smaller than the first element. If it satisfies the condition, the second element's position is inserted to the first element before the value with a higher value. ([4, 2, 16, 8] -> [2, 4, 16, 8]) Next, a similar process is done for the third element by comparing this element to the values of the first two elements in terms of their value. Again, if it satisfies the condition, the third element's position is inserted into the correct position. This process repeats until the entire list is sorted.
([2, 4, 16, 8] -> [2, 4, 16, 8] -> [2, 4, 8, 16])

Bubble Sort
You compare neighboring items two at a time starting from the beginning. If the left item is bigger than the right item, you swap them. As you sweep through the list, the largest value gradually "bubbles up" to the very end. You repeat this process from the start until you can sweep through the whole list without making a single swap. For example, a list [4, 2, 7, 1], compare 4 and swap them -> [2, 4, 7, 1]. Compare 4 and 7 (no swap). Compare 7 and 1 and swap them -> [2, 4, 1, 7]. In just one pass, the largest number (7) bubbled to the very end.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Extent of Use:

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